DEMO-001 — Submission

LaTeX response — synthetic student

Q1 — Calculus

Let $u = x^2$, so $\mathrm{d}u = 2x\,\mathrm{d}x$. Therefore

$$I = \frac{1}{2}\int_0^1 e^u\,\mathrm{d}u = \frac{1}{2}\left[e^u\right]_0^1 = \frac{e-1}{2}.$$

Q2 — Mechanics

The normal force is $N = mg\cos 30^\circ$. Along the slope,

$$ma = mg\sin 30^\circ - \mu_k N, \qquad a = 9.81(\sin 30^\circ - 0.20\cos 30^\circ) = 3.20\,\mathrm{m\,s^{-2}}.$$

Starting from rest, $v^2 = 2as$, so

$$v = \sqrt{2(3.20)(3)} = 4.38\,\mathrm{m\,s^{-1}}.$$

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Q3 — Linear algebra

If $Ax = 0$, then $A^\top Ax = 0$. Conversely, if $A^\top Ax = 0$, left-multiplying by $x^\top$ gives

$$x^\top A^\top Ax = \|Ax\|^2 = 0,$$

hence $Ax = 0$. The kernels are equal, so rank–nullity gives equal ranks.

Q4 — Programming

```
import math

def stable_softmax(xs):
    if not xs:
        return []
    exps = [math.exp(x) for x in xs]
    total = sum(exps)
    return [value / total for value in exps]
```